

Breaking the Power Wall: Chiplet-Based Power Management Architectures for HPC 2.5D/3D Processors

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Abstract—As High-Performance Computing (HPC) processors transition into the post-Moore era, power management has emerged as a primary bottleneck, often referred to as the “Power Wall.” This paper explores advanced power management architectures, specifically focusing on chiplet-based integrated voltage regulators (IVRs) and Vertical Power Delivery (VPD) to sustain the increasing current demands of 2.5D and 3D integrated circuits. We discuss the transition from conventional lateral power delivery to high-density, vertically integrated solutions that reduce parasitic losses and improve transient response. The proposed Multi-Stage Modular Switched Capacitor Voltage Regulator (MMSCVR) and Regulated Resonant Switched-capacitor (R^2SC) converters demonstrate significant improvements in power density and efficiency, reaching up to 1339 W/in^3 and 95.6% peak efficiency, respectively.

Index Terms—Power Management, Chiplets, Vertical Power Delivery (VPD), Integrated Voltage Regulator (IVR), Switched-Capacitor Converters, HPC Processors.

I. INTRODUCTION

The continuous scaling of HPC processors has led to a dramatic increase in Thermal Design Power (TDP), with high-power GPUs now exceeding 500W-700W and approaching kilowatt levels[cite: 1, 2, 42]. This trend, coupled with decreasing supply voltages (V_{DD}), has created a “Power Wall” where conventional power delivery architectures can no longer sustain the required current densities[cite: 1, 3]. Traditional lateral power delivery (LPD) faces severe limitations due to high I^2R losses in the “last mile” of the power distribution network (PDN) and limited PCB copper thickness[cite: 4, 19]. To unlock the full performance of next-generation XPU, power delivery must transition to a new paradigm where integrated voltage regulators are placed in close proximity to the compute dies, utilizing advanced 2.5D and 3D packaging technologies[cite: 3, 7].

II. MODULATOR DESIGN AND IMPLEMENTATION

To address these challenges, several innovative modulator designs are proposed:

- **MMSCVR in 16nm CMOS:** A Multi-Stage Modular Switched Capacitor Voltage Regulator (MMSCVR) is designed as a base-die IVR. It utilizes high-density on-chip MIM capacitors and a Dickson topology to achieve excellent scalability for both input voltage (V_{IN}) and output current (I_{OUT})[cite: 22, 23, 28].
- **Flying Capacitor IVR (FCIVR) Chiplet:** Fabricated in 65nm CMOS, this chiplet is designed for heterogeneous integration. It features an autonomous handshake driving strategy to manage complex switch networks and optimize deadtime for improved Power Conversion Efficiency (PCE)[cite: 14, 17].

- **R^2SC Converter:** A Regulated Resonant Switched-capacitor converter is implemented for 48V-to-1V applications. It employs a Unified Hard-Soft Switching Control (UHSC) strategy, combining PWM and PFM to ensure Zero Crossing Detection (ZCD) for Zero Current Switching (ZCS), thereby maximizing efficiency across a wide input range (30-60V)[cite: 35, 36, 40].

III. EXPERIMENTAL RESULTS AND SYSTEM DEMONSTRATION

The experimental results highlight significant performance leaps over state-of-the-art solutions:

- **Vertical Power Delivery:** The FCIVR VPD chiplet achieved a measured current density of 3.2 A/mm^2 with a fast load transient figure of merit (FoM) of 32ns[cite: 18].
- **Monolithic Efficiency:** The 16nm MMSCVR provides a fully monolithic solution that eliminates the need for bulky package inductors, thereby reducing Z-height and increasing pin availability for data bandwidth[cite: 21, 30].
- **High Density R^2SC :** The R^2SC converter prototype demonstrated a peak efficiency of 95.6% and a power density of 1339 W/in^3 . Thermal imaging confirmed the effectiveness of the UHSC strategy in optimizing frequency and reducing heat dissipation under fan cooling[cite: 38, 39].

IV. CONCLUSION

The integration of voltage regulators into chiplets and the adoption of Vertical Power Delivery are essential to overcome the HPC Power Wall[cite: 42]. By utilizing advanced nodes like 16nm and innovative topologies such as the MMSCVR and R^2SC , power management systems can achieve the high current densities and fast transient responses required by modern DVFS and Turbo Boost technologies[cite: 5, 30]. Future developments will continue to leverage interdisciplinary frontiers in packaging, device materials, and control architectures to sustain the performance scaling of 2.5D and 3D processor architectures[cite: 7, 42].